

Q9. The circuit below is a common source amplifier with a resistor load. The parasitic capacitors are considered in the amplifier bandwidth calculation.

(a) Identify the number of poles and zeros in this circuit (2 points)

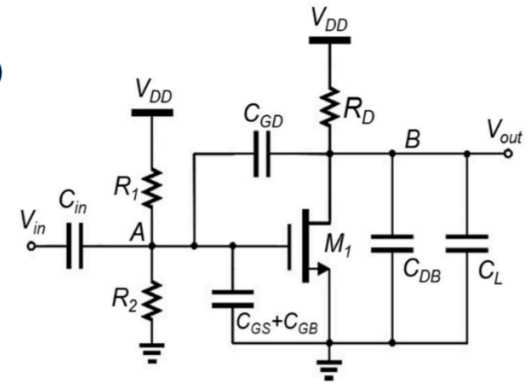
Number of poles: 2

$$\text{Values of poles } \omega_1 = \frac{1}{(R_1 || R_2)[C_{GD}(1 + g_{m1}R_D) + C_{GS} + C_{GB} + C_{in}]}$$

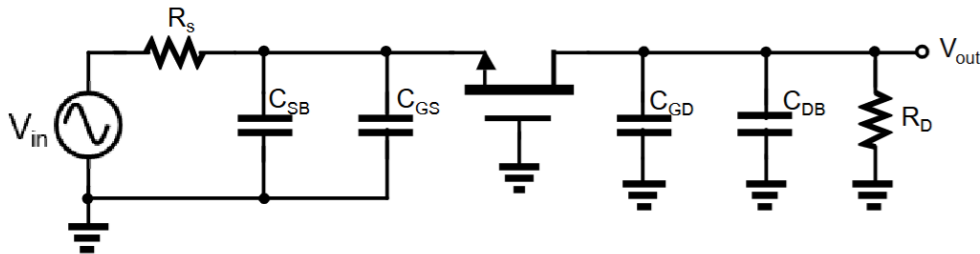
$$\omega_2 = \frac{1}{R_D \left[C_{GD} \left(1 + \frac{1}{g_{m1}R_D} \right) + C_{DB} + C_L \right]}$$

Number of zeros: 2

Values of zeros: $\omega_1=0$, $\omega_2=g_{m1}/C_{GD}$



Apply Association of Poles with Nodes

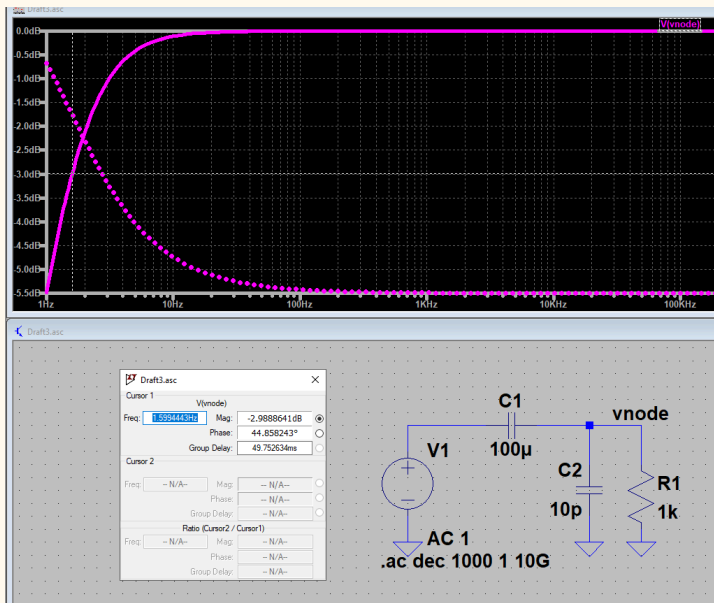


- Since all capacitors go to ground from respective nodes, finding poles reduces to finding R_{eq} and C_{eq} at those nodes. Each pole is associated with a node.
- R_{eq} is the equivalent resistance seen at the node when **all capacitors are open circuit**.
- C_{eq} is the equivalent capacitance seen at the node when **all resistors are open circuit**.

$$R_{eq1} \approx R_D \quad C_{eq1} = C_{GD} + C_{DB} \rightarrow \omega_{p1} = \frac{1}{R_{eq1}C_{eq1}} = \frac{1}{R_D(C_{GD} + C_{DB})} = \frac{1}{R_D C_D}$$

$$R_{eq2} \approx R_s || \frac{1}{g_m + g_{mb}} \quad C_{eq2} = C_{GS} + C_{SB} \rightarrow \omega_{p2} = \frac{1}{R_{eq2}C_{eq2}} = \frac{1}{\left(R_s || \frac{1}{g_m + g_{mb}} \right) C_S}$$

- - the source need to be shorted and the load connected when finding poles
 - so there is always no poles when standing at Vin
 - standing at vout there is poles, because its connected to load
 - so there is only zero due to Cin
 - the pole at node A will include Cin, since Cin is to ground
 - so in ω_1 should + Cin
 - this is the pole that give the HPF corner
 - this pole is at lower frequency than the pole at output, the lower this pole the wider the bandwidth
 - but the dominant pole that limits bandwidth on the upper frequency cutoff is still the output pole
- 2 poles 2 zero is correct



$$\frac{1}{2\pi RC}$$

$$= 1.59154943092$$

$$R = 1000$$

$$-10 \text{ } \dots \text{ } 1000$$

$$C = 100 \cdot 10^{-6}$$

$$= 0.0001$$

- this shows the HPF input giving 1 zero and 1 pole
- There is no extra pole from the HPF stage in front, its combined with the node A one

Q.9 The circuit below is a common source amplifier with a resistor load. The parasitic capacitors are considered in the amplifier bandwidth calculation.

(a) Identify the number of poles and zeros in this circuit. (2 points)

Number of poles: 2 Values of poles: $\omega_{p1} = \frac{1}{(R_1 || R_2)(C_{GS} + C_{GB} + C_{GD}(1 - A_v) + C_{in})}$

Number of zeros: 1 Values of zeros: $\omega_{z1} = 0$ $\omega_{z2} = \frac{1}{C_{GD}(R_1 || R_2)}$

(b) If the dominant pole is at the output node B, if C_L increases from 2pF to 5pF, amplifier's bandwidth decreases / increases / remains constant (delete as applicable). (1 point)

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$\frac{1}{RC}$

only wrong is the zero